

Homework 7: Preliminaries to Scattering Theory



HW 7.1: In this question we will look into the state $i \rightarrow i$ transitions in time-dependent perturbation theory. It contains a lot of familiar results that nicely wraps up the perturbation theory chapter.

Working in the interaction picture,

$$|\psi_I(t)\rangle = e^{iH_0 t} |\psi(t)\rangle = \sum_n c_n(t) |n\rangle, \quad c_i(0) = 1, \quad c_{n \neq i}(0) = 0, \quad (1)$$

the Dyson expansion can be written as

$$c_n(t) = \delta_{ni} + c_n^{(1)}(t) + c_n^{(2)}(t) + \dots \quad (2)$$

with

$$c_n^{(1)}(t) = -i \int_0^t dt_1 e^{i(E_n - E_i)t_1} V_{ni}, \quad (3)$$

$$c_n^{(2)}(t) = - \sum_m \int_0^t dt_1 \int_0^{t_1} dt_2 e^{i(E_n - E_m)t_1} e^{i(E_m - E_i)t_2} V_{nm} V_{mi}. \quad (4)$$

a) Show that to first order, the correction to the initial-state amplitude is just a phase shift and that the energy of the state is corrected just as in time-independent perturbation theory.

b) Starting from the Dyson expression with $n = i$, show that

$$c_i^{(2)}(t) = - \sum_m |V_{mi}|^2 \int_0^t dt_1 \int_0^{t_1} dt_2 e^{i(E_i - E_m)(t_1 - t_2)}. \quad (5)$$

What does the term with $m = i$ correspond to (see part (a))? For the remaining $m \neq i$ terms, perform the time integrals and identify the term that grows linearly with t . Show that the coefficient of this “secular” term reproduces the second-order energy shift of time-independent perturbation theory.

c) We now want to consider the case where the states $|m\rangle$ include a continuum of energies. In general, the energy denominator in part (b) must be regulated. One physical way to do this is to adiabatically switch on the perturbation from the infinite past, replacing $V \rightarrow V e^{\eta t}$ with $\eta \rightarrow 0^+$ (this is the approach in Sakurai’s book). Redo the calculation in part (b) with this replacement. You should find that the net effect on the energy denominator is just

$$\frac{1}{E_i - E_m} \rightarrow \frac{1}{E_i - E_m + i\epsilon}, \quad \epsilon = \eta \rightarrow 0^+. \quad (6)$$

Hint: Each V_I insertion now carries an extra factor of $e^{\eta t_k}$. The secular (linear-in- t) piece of $c_i^{(2)}(t)$ should look like

$$c_i^{(2)}(t) \supset -i t \sum_{m \neq i} \frac{|V_{mi}|^2}{E_i - E_m + i\epsilon}. \quad (7)$$

d) Now replace the sum over intermediate states by an integral weighted by a density of states $\rho(E)$. Use the identity,

$$\frac{1}{x + i\epsilon} = \mathcal{P}\left(\frac{1}{x}\right) - i\pi \delta(x), \quad (8)$$

where $\mathcal{P}$ denotes the Cauchy principal value, to separate the energy shift into a real and an imaginary part. Show that, for large t ,

$$c_i(t) \simeq 1 - i \Delta_i t, \quad \Delta_i = \Delta E_i - \frac{i}{2} \Gamma_i, \quad (9)$$

with

$$\Delta E_i = V_{ii} + \mathcal{P} \int dE \frac{|V_{iE}|^2 \rho(E)}{E_i - E}, \quad (10)$$

$$\Gamma_i = 2\pi \int dE |V_{iE}|^2 \rho(E) \delta(E_i - E) = 2\pi |V_{iE_i}|^2 \rho(E_i). \quad (11)$$

What are the physical interpretations of ΔE_i and Γ_i ? In particular, discuss what happens to the state $|i\rangle$ if $\Gamma_i \neq 0$.



The results above show that an unstable state acquires a complex energy $\mathcal{E}_i = E_i + \Delta E_i - i\Gamma_i/2$. This has direct consequences for scattering and transition processes that proceed through $|i\rangle$ as an intermediate state.



Why integrate from $t = 0$ and not $t = -\infty$ as in Sakurai's approach? Our Dyson series was derived by integrating Schrödinger's equation from $t = 0$, so this is the only consistent way to do the calculation without rederiving it. My approach follows: [Hitoshi Murayama's lecture notes](#).

Solutions



HW 7.1:

Let us first look into the claims of the setup of the question. This is just a little different picture for the Dyson series than what we had been working with, so let us derive it. Recall that in the interaction picture, Schrödinger's equation reads

$$i \frac{d}{dt} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle, \quad V_I(t) = e^{iH_0 t} V e^{-iH_0 t}, \quad (12)$$

and therefore

$$|\psi_I(t)\rangle = U_I(t, 0) |i\rangle. \quad (13)$$

We have already derived the Dyson expansion for $U_I(t, 0)$:

$$U_I(t, 0) = 1 + (-i) \int_0^t dt_1 V_I(t_1) + (-i)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 V_I(t_1) V_I(t_2) + \dots \quad (14)$$

Expanding $|\psi_I(t)\rangle$ in the H_0 eigenbasis,

$$|\psi_I(t)\rangle = \sum_n c_n(t) |n\rangle, \quad c_n(t) = \langle n | \psi_I(t) \rangle = \langle n | U_I(t, 0) | i \rangle, \quad (15)$$

and comparin this with the Dyson expansion, we obtain the relations in the setup of the question in terms of interaction picture matrix elements of V . For example,

$$c_n(t) = \delta_{ni} + c_n^{(1)}(t) + c_n^{(2)}(t) + \dots, \quad (16)$$

where

$$c_n^{(1)}(t) = -i \int_0^t dt_1 \langle n | V_I(t_1) | i \rangle. \quad (17)$$

By the definition of $V_I(t)$ and the Hermiticity of H_0 , we have $\langle n | V_I(t) | m \rangle = e^{i(E_n - E_m)t} V_{nm}$, so

$$c_n^{(1)}(t) = -i \int_0^t dt_1 e^{i(E_n - E_i)t_1} V_{ni} = -i \int_0^t dt_1 e^{i\omega_{ni}t_1} V_{ni}. \quad (18)$$

and similarly for $c_n^{(2)}(t)$.

a) Investigating the $n = i$ case, we get

$$c_i^{(1)}(t) = -i \int_0^t dt_1 e^{i(E_i - E_i)t_1} V_{ii} = -i V_{ii} \int_0^t dt_1 = -i V_{ii} t. \quad (19)$$

Therefore, to first order,

$$c_i(t) = c_i^{(0)}(t) + c_i^{(1)}(t) = 1 - i V_{ii} t \simeq e^{-i V_{ii} t}. \quad (20)$$

Therefore, at first order, the $i \rightarrow i$ piece of the perturbation is equivalent to a phase shift of the initial-state amplitude.

In the Schrödinger picture, the i -th component of the state $|\psi(t)\rangle$ is $e^{-iE_i t} c_i(t)$, so

$$e^{-iE_i t} c_i(t) \simeq e^{-i(E_i + V_{ii})t}, \quad (21)$$

recovering the time-independent perturbation theory result

$$E_i \simeq E_i^{(0)} + E_i^{(1)} = E_i^{(0)} + V_{ii}. \quad (22)$$

b) Again setting $n = i$:

$$c_i^{(2)}(t) = - \sum_m |V_{mi}|^2 \int_0^t dt_1 \int_0^{t_1} dt_2 e^{i(E_i - E_m)(t_1 - t_2)}, \quad (23)$$

where $V_{im}V_{mi} = |V_{mi}|^2$. The $m = i$ term is just

$$-|V_{ii}|^2 \int_0^t dt_1 \int_0^{t_1} dt_2 = -\frac{|V_{ii}|^2 t^2}{2} = \frac{1}{2!}(-iV_{ii}t)^2, \quad (24)$$

which is the second-order piece of the exponentiation $e^{-iV_{ii}t}$ from part (a).

For $m \neq i$, we perform the integrals as usual

$$\int_0^{t_1} dt_2 e^{-i\omega_{im}t_2} = \frac{1 - e^{-i\omega_{im}t_1}}{i\omega_{im}}, \quad (25)$$

such that

$$c_i^{(2)}(t) \supset - \sum_{m \neq i} |V_{mi}|^2 \left[\frac{it}{\omega_{im}} + \frac{1 - e^{i\omega_{im}t}}{\omega_{im}^2} \right], \quad (26)$$

where $\supset$ means “contains” and we focused on the $m \neq i$ terms. The first term grows linearly with t . This is the secular term. The second is bounded and oscillatory. Keeping only the secular term, we have

$$c_i^{(2)}(t)|_{\text{secular}} = -it \sum_{m \neq i} \frac{|V_{mi}|^2}{E_i - E_m}. \quad (27)$$

Again, comparing with the structure $c_i(t) \simeq 1 - i\Delta_i^{(2)}t$ gives

$$\Delta_i^{(2)} = \sum_{m \neq i} \frac{|V_{mi}|^2}{E_i - E_m}, \quad (28)$$

which is the standard result from time-independent second-order perturbation theory for the energy of the state.



Just as before, this too is a phase shift and does not affect the probability $|c_i(t)|^2$ at this order. The point here is that the $i \rightarrow i$ pieces of time-dependent perturbation theory contain the time-independent perturbation theory energy corrections. The non-secular oscillatory pieces are not captured by time-independent perturbation theory, but we never expected them to be.

c) The replacement $V \rightarrow Ve^{\eta t}$ means that for each insertion of V_I in the Dyson series, we get an extra factor of $e^{\eta t_k}$. For example, the second-order diagonal amplitude becomes

$$c_i^{(2)}(t) = - \sum_{m \neq i} |V_{mi}|^2 \int_0^t dt_1 \int_0^{t_1} dt_2 e^{\eta t_1} e^{i\omega_{mi}t_1} e^{\eta t_2} e^{-i\omega_{mi}t_2}, \quad (29)$$

with $\omega_{mi} = E_i - E_m$ as before. The inner integral is

$$\int_0^{t_1} dt_2 e^{(\eta - i\omega_{mi})t_2} = \frac{e^{(\eta - i\omega_{mi})t_1} - 1}{\eta - i\omega_{mi}}. \quad (30)$$

Substituting and performing the outer integral, the term that grows linearly with t arises when the oscillatory factor in the numerator cancels the one in the exponential of the outer integrand. Concretely, the first term in the numerator gives

$$- \sum_{m \neq i} \frac{|V_{mi}|^2}{\eta - i\omega_{im}} \int_0^t dt_1 e^{2\eta t_1} = - \sum_{m \neq i} \frac{|V_{mi}|^2}{\eta - i\omega_{im}} \frac{e^{2\eta t} - 1}{2\eta}. \quad (31)$$

For small η , $(e^{2\eta t} - 1)/2\eta \rightarrow t$, while $1/(\eta - i\omega_{im}) \rightarrow i/\omega_{im}$ recovers the unregulated result. Instead, we keep $\epsilon = \eta \rightarrow 0^+$ terms in the denominator to find

$$\frac{1}{\eta - i\omega_{im}} = \frac{1}{-i(\omega_{im} + i\eta)} = \frac{i}{\omega_{im} + i\epsilon}, \quad (32)$$

so

$$c_i^{(2)}(t)|_{\text{secular}} = -it \sum_{m \neq i} \frac{|V_{mi}|^2}{E_i - E_m + i\epsilon}, \quad (33)$$

as expected.



You can also recover this result by integrating from the infinite past as in Sakurai's book. In both approaches one must decide how to keep the expressions regular when $E_m \rightarrow E_i$. The adiabatic switching approach gives a physical interpretation to the $i\epsilon$ prescription: the perturbation was not sizable until $t \simeq -1/\epsilon$, so the system had no reason to respond to it until then. In our approach of integrating from 0 the $i\epsilon$ prescription is just a convenient way to keep the integrals well-defined, but it does not have a direct physical interpretation.

d) Now we manipulate this result into the form of a real and imaginary part for the phase shift. We also consider the continuous limit for the energy spectrum. This is a simple replacement of

$$\sum_{m \neq i} \rightarrow \int dE \rho(E), \quad (34)$$

Using the relation provided with the Cauchy principal value, the secular part of $c_i(t)$ becomes (including now the first-order piece from (a))

$$c_i(t) \simeq 1 - i \left[V_{ii} + \mathcal{P} \int dE \frac{|V_{iE}|^2 \rho(E)}{E_i - E} - i\pi \int dE |V_{iE}|^2 \rho(E) \delta(E_i - E) \right] t. \quad (35)$$

Comparing this to the form $c_i(t) \simeq 1 - i \Delta_i t$ with

$$\Delta_i = \Delta E_i - i\Gamma_i/2, \quad (36)$$

we read off

$$\Delta E_i = V_{ii} + \mathcal{P} \int dE \frac{|V_{iE}|^2 \rho(E)}{E_i - E} \quad (37)$$

and

$$\Gamma_i = 2\pi |V_{iE_i}|^2 \rho(E_i). \quad (38)$$

i

The **real part** $\text{Re}(\Delta_i) = \Delta E_i$ is the energy shift of the level $|i\rangle$. Its first term is the first-order correction; its second term is the (principal-value regulated) second-order correction from time-independent perturbation theory.

The **imaginary part** $\text{Im}(\Delta_i) = -\Gamma_i/2$ is the total decay rate out of the state $|i\rangle$, identical to Fermi's Golden Rule summed over all accessible final states. If $\Gamma_i \neq 0$ (which requires a continuum of states at energy E_i), the state is said to be *unstable*. The secular terms resum into an exponential, so that to second order in perturbation theory,

$$c_i(t) \simeq e^{-i\Delta_i t} = e^{-i\Delta E_i t} e^{-\Gamma_i t/2}, \quad (39)$$

which means that not only the states acquire an energy shift, but also the probability of finding the system in the state $|i\rangle$ decays exponentially with time:

$$|c_i(t)|^2 \simeq e^{-\Gamma_i t}. \quad (40)$$

Note how the factor of 1/2 in the definition of Γ in Eq. (36) makes sure that the decay rate of the probability is Γ_i rather than $2\Gamma_i$.

So, the perturbation promotes the energy of our state to a complex quantity,

$$E_i \rightarrow \mathcal{E}_i = E_i + \Delta E_i - \frac{i}{2} \Gamma_i. \quad (41)$$

The real and imaginary parts of $\mathcal{E}_i$ are calculable from 1) the matrix elements of the perturbation and 2) the density of states at the energy of the state.

i

Causality and the $i\epsilon$ prescription. The $i\epsilon$ prescription in the energy denominator is a consequence of the adiabatic switching on of the perturbation from the infinite past. The structure of the energy denominator

$$\frac{1}{E_i - E_m + i\epsilon}, \quad (42)$$

is said to have a “pole” in the complex energy plane at $E_m = E_i - i\epsilon$ (the pole is infinitesimally *below* the real axis). Imagine instead that $\eta \rightarrow -\eta$ in the adiabatic switching, so that the perturbation is switched on from the infinite future. In that case, the $i\epsilon$ prescription would be flipped, and the energy denominator would be $1/(E_i - E_m - i\epsilon)$, showing a pole $E_m = E_i + i\epsilon$ (the pole is infinitesimally *above* the real axis). This solution would make sense if the particle was propagating backwards in time, but it is not physical in quantum mechanics. This is connected to our discussion of causality with propagators and the different boundary conditions: the retarded propagator corresponds to the $i\epsilon$ prescription, while the advanced propagator corresponds to the flipped $-i\epsilon$ prescription. We will return to the propagator language and see these prescriptions arise from the Fourier transform of the retarded and advanced Green's functions.



In relativistic quantum field theory, the $i\epsilon$ prescription also shows up in the “Feynman propagator”:

$$\frac{1}{p^2 - m^2 + i\epsilon}, \quad (43)$$

where $p^2 = p_\mu p^\mu = (p^0)^2 - \mathbf{p}^2$ is the four-momentum squared and m is the mass of the particle. If we rewrite this expression as

$$\frac{1}{(p^0)^2 - \mathbf{p}^2 - m^2 + i\epsilon} = \frac{1}{(p^0)^2 - E_{\mathbf{p}}^2 + i\epsilon}, \quad (44)$$

where we defined the on-shell energy $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} > 0$. We see two poles in the complex p^0 plane:

$$p^0 = \pm \sqrt{E_{\mathbf{p}}^2 - i\epsilon} \simeq \pm E_{\mathbf{p}} \mp i \frac{\epsilon}{2E_{\mathbf{p}}}. \quad (45)$$

One corresponds to positive energy particles propagating forward in time ($-i\epsilon$, pole below the real axis), while the other corresponds to negative energy particles propagating backward in time ($+i\epsilon$, pole above the real axis). So, in QFT, the $i\epsilon$ prescription is mixed inside the propagator and it accommodates both positive and negative energy solutions, which correspond to particles and antiparticles, respectively.